

Rabin BK

Machine Learning Researcher | M.Sc. in Data Science at FAU
Erlangen, Bavaria – Germany

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Education

Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU)

M.Sc. Data Science

Erlangen, Germany
2023 – Present

- **Thesis:** *Structured Extraction of Retrospective MIMIC-IV Studies* | [Code](#)
- **Supervisor:** Prof. Dr. David Benjamin Blumenthal, Dr. Christel Sirrochi
- **Focus:** RAG system and evaluation, Large Language Models (LLM), Retrospective Study

Tribhuvan University

B.Sc. Computer Science and Information Technology, grade: 73.98/100 (First Division)

Kathmandu, Nepal
2016 – 2020

- **Research Project:** *Nepali Number Plate Recognition*
- **Focus:** Image processing and neural networks

Experience

Primetals Technologies Germany GmbH

Werkstudent – Data Science / Machine Learning

Erlangen, Germany
December 2024 –
Present

- Evaluated 35+ open-source time-series datasets with different sampling rates to support benchmarking of industrial anomaly-detection workflows.
- Built Python workflows for sensor-data preprocessing, labeling, model training, and evaluation on an internal industrial analytics platform.
- Analyzed raw and aggregated time-series features, identifying effective feature combinations for anomaly detection.
- Designed a two-stage motif-search (using STUMPY) approach using symbolic time-series representation (SAX) for fast candidate filtering, followed by exact subsequence matching for high-resolution sensor data.

Forschungszentrum Jülich

Werkstudent

Nuremberg, Germany
May 2024 – January
2025

- Developed an **inventory and privilege management system** for experiments at HIERN (Helmholtz Institute Erlangen-Nürnberg), enabling controlled user access with automated database backup using **automysqlbackup**.
- Implemented Python scripts to connect JupyterHub to the MySQL database, allowing users to submit, replicate, and track experiments.

Lumbini Academy College

Computer Lecturer

Kathmandu, Nepal
December 2021 –
January 2023

- Delivered lectures on programming, algorithms, and database fundamentals
- Supervised student projects, guided practical lab sessions
- Trained **150+ students** across **3 annual shifts**, with **25+ students per batch**

Wiseyak Inc.

Backend Intern

Kathmandu, Nepal
July 2021 – Oct 2021

- Worked with GraphQL queries to query data for the dashboard and report related to FHIR (Fast Healthcare Interoperability Resources).
- Extracted data from PDFs (using pdfplumber), created network graphs and prepared a report.

Recent Projects

- **Thesis Work: Structured Extraction of Retrospective MIMIC-IV Studies** (Ongoing): Designing and developing a RAG-based clinical study analysis pipeline using locally-hosted LLMs (Ollama), Qdrant vector search with BM25 hybrid retrieval, and evaluating system performance using the RAGAs framework, as part of an ongoing Master's thesis.
- **Investigating Informative Missingness in Time Series Laboratory Data** | **Paper** (2025): Developed and containerized a modular ML pipeline (XGBoost, Random Forest, CatBoost, Gradient Boosting, Logistic Regression) to predict Missingness Not at Random (MNAR) in the MIMIC-IV database across Aplasia and Neutropenic Fever cohorts, deploying reproducibly via Docker and Apptainer/HPC as part of a 10 ECTS Bioinformatics lab project.
- **Yugantar – Internal Fund Management System** (2025 - Present): Building a full-stack, policy-driven cooperative management platform (FastAPI, PostgreSQL/Alembic, React, Docker) to handle role-based deposits, loans, and administrative workflows, with ongoing development toward more advanced features.
- **Gele Trekking** (2025 - Present): Mentored a student in designing the database schema and architecture for a MERN-stack trekking company web application, guiding backend/frontend integration and Docker-based deployment, while personally driving further development ("vibe coding") through to the current deployment phase. | [Live Website](#)

Skills

Programming

Python (advanced), SQL, Bash

LLMs & RAG

Ollama, Qdrant, BM25, Hybrid Search, RAGAs, Prompt Engineering

Machine Learning

PyTorch, TensorFlow, Scikit-learn, HuggingFace, OpenCV

ML Systems

Docker, Git/GitHub, FastAPI, Flask, Streamlit, Firestore, GitHub Actions, GitLab CI/CD

Languages

English (C1), German (B1), Nepali (Native)